Double-Angle Formulas and Trigonometric Equations

ANSWER KEY

Section 1: Double-Angle Formulas

1 State the double-angle formula for $\sin(2\theta)$.

a. $2\sin\theta\cos\theta$ ✓

b. $\cos^2\theta - \sin^2\theta$

c. $2\sin\theta$

d. $\sin^2\theta$

$\sin(2\theta) = \sin(\theta+\theta) = \sin\theta\cos\theta + \cos\theta\sin\theta = 2\sin\theta\cos\theta$.

2 Which of the following is NOT an equivalent form of $\cos(2\theta)$?

a. $1 - 2\cos^2\theta$ ✓

b. $\cos^2\theta - \sin^2\theta$

c. $2\cos^2\theta - 1$

d. $1 - 2\sin^2\theta$

The correct forms are $\cos^2\theta - \sin^2\theta$, $2\cos^2\theta - 1$, and $1 - 2\sin^2\theta$. $1 - 2\cos^2\theta = -\cos(2\theta)$.

3 If $\cos\theta = 4/5$ with θ in Quadrant IV, find $\sin(2\theta)$.

a. $-24/25$ ✓

b. $24/25$

c. $-7/25$

d. $7/25$

In QIV, $\sin\theta = -3/5$. $\sin(2\theta) = 2\sin\theta\cos\theta = 2(-3/5)(4/5) = -24/25$.

Section 2: Half-Angle and Power-Reducing Formulas

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4 State the power-reducing formula for $\sin^2 x$.

a. $\frac{1-\cos(2x)}{2}$ ✓

b. $\frac{1+\cos(2x)}{2}$

c. $\frac{1-\sin(2x)}{2}$

d. $1-\cos^2 x$

From $\cos(2x) = 1 - 2\sin^2 x$, solving for $\sin^2 x$ gives $\sin^2 x = \frac{1-\cos(2x)}{2}$.

5 Find the exact value of $\cos(22.5^\circ)$ using the half-angle formula $\cos(\theta/2) = \sqrt{\frac{1+\cos\theta}{2}}$.

a. $\frac{\sqrt{2+\sqrt{2}}}{2}$ ✓

b. $\frac{\sqrt{2-\sqrt{2}}}{2}$

c. $\frac{\sqrt{2}+1}{2}$

d. $\frac{\sqrt{3}+1}{2}$

$\cos(45^\circ/2) = \sqrt{\frac{1+\sqrt{2}/2}{2}} = \sqrt{\frac{2+\sqrt{2}}{4}} = \frac{\sqrt{2+\sqrt{2}}}{2}$.

6 Rewrite $\cos^4 x$ in terms of first powers of cosines of multiple angles.

a. $\frac{3}{8} + \frac{1}{2}\cos(2x) + \frac{1}{8}\cos(4x)$ ✓

b. $\frac{1}{4}(1+\cos(2x))^2$

c. $\frac{1}{2} + \frac{1}{2}\cos(4x)$

d. $\frac{3}{8} - \frac{1}{2}\cos(2x)$

$\left(\frac{1+\cos 2x}{2}\right)^2 = \frac{1+2\cos 2x+\cos^2 2x}{4} = \frac{1+2\cos 2x+(1+\cos 4x)/2}{4} = \frac{3}{8} + \frac{1}{2}\cos(2x) + \frac{1}{8}\cos(4x)$.

Section 3: Solving Linear Trigonometric Equations

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7 Solve $2\sin x - 1 = 0$ on the interval $[0, 2\pi)$.

a. $\frac{\pi}{6}$ **and** $\frac{5\pi}{6}$ ✓

b. $\frac{\pi}{3}$ and $\frac{2\pi}{3}$

c. $\frac{\pi}{6}$ and $\frac{11\pi}{6}$

d. $\frac{5\pi}{6}$ and $\frac{7\pi}{6}$

$\sin x = 1/2$. On $[0, 2\pi)$, sine is positive in Quadrants I and II at $x = \pi/6$ and $5\pi/6$.

8 Solve $\sqrt{3}\tan x + 1 = 0$ on $[0, 2\pi)$.

a. $\frac{5\pi}{6}$ **and** $\frac{11\pi}{6}$ ✓

b. $\frac{2\pi}{3}$ and $\frac{5\pi}{3}$

c. $\frac{\pi}{6}$ and $\frac{7\pi}{6}$

d. $\frac{5\pi}{6}$ and $\frac{7\pi}{6}$

$\tan x = -1/\sqrt{3}$. Tangent is negative in Quadrants II and IV at $x = 5\pi/6$ and $11\pi/6$.

9 Find all solutions to $\cos(2x) = 0$ on $[0, \pi)$.

a. $\frac{\pi}{4}$ **and** $\frac{3\pi}{4}$ ✓

b. $\frac{\pi}{2}$

c. $\frac{\pi}{4}$ only

d. $\frac{\pi}{2}$ and $\frac{3\pi}{2}$

$2x = \frac{\pi}{2} + k\pi \Rightarrow x = \frac{\pi}{4} + \frac{k\pi}{2}$. On $[0, \pi)$, solutions are $\pi/4$ and $3\pi/4$.

Section 4: Quadratic-Type and Multiple-Angle Equations

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10 Solve $2\sin^2 x - \sin x - 1 = 0$ on $[0, 2\pi)$.

a. $\frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$ ✓

b. $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$

c. $\frac{\pi}{2}, \frac{5\pi}{6}$

d. $\frac{\pi}{3}, \frac{5\pi}{3}$

Factor: $(2\sin x + 1)(\sin x - 1) = 0$. $\sin x = 1 \Rightarrow x = \pi/2$. $\sin x = -1/2 \Rightarrow x = 7\pi/6, 11\pi/6$.

11 Solve $\sin(2x) = \cos x$ on $[0, 2\pi)$.

a. $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$ ✓

b. $\frac{\pi}{6}, \frac{5\pi}{6}$

c. $\frac{\pi}{3}, \frac{2\pi}{3}$

d. $\frac{\pi}{2}, \frac{3\pi}{2}$

$2\sin x\cos x - \cos x = 0 \Rightarrow \cos x(2\sin x - 1) = 0$. $\cos x = 0 \Rightarrow x = \pi/2, 3\pi/2$.
$\sin x = 1/2 \Rightarrow x = \pi/6, 5\pi/6$.

12 Solve $\cos(2x) + \cos x = 0$ on $[0, 2\pi)$.

a. $\frac{\pi}{3}, \pi, \frac{5\pi}{3}$ ✓

b. $\frac{\pi}{6}, \pi, \frac{11\pi}{6}$

c. $\frac{\pi}{2}, \frac{3\pi}{2}$

d. $\frac{\pi}{3}, \frac{2\pi}{3}$

$2\cos^2 x - 1 + \cos x = 0 \Rightarrow (2\cos x - 1)(\cos x + 1) = 0$. $\cos x = 1/2 \Rightarrow x = \pi/3, 5\pi/3$.
$\cos x = -1 \Rightarrow x = \pi$.

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